MATSYA: An Autonomous Biomimetic Underwater Robot for Low-Disturbance Ecological Monitoring

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Abstract. Underwater monitoring platforms are critical for environmental observation and aquatic inspection. Conventional propeller-driven systems, however, introduce significant trade-offs regarding energy efficiency, manoeuvrability, and disturbance to aquatic ecosystems. To address these limitations, this paper introduces MATSYA, an autonomous biomimetic underwater robot inspired by fish locomotion [4,5]. MATSYA provides a comprehensive framework that integrates fish-inspired propulsion, computational fluid dynamics (CFD), physics-informed neural networks (PINNs), ROS 2 middleware, and closed-loop control. CFD is utilized to evaluate body and fin geometries, estimating flow fields, thrust, drag, and wake characteristics [2,3]. Furthermore, PINNs are investigated as surrogate models to accelerate specific flow predictions without entirely replacing high-fidelity CFD [1]. The system's software architecture relies on ROS 2 for sensor integration, state estimation, navigation, and mission-level control. Through a structured, phased development workflow ranging from numerical simulation to field validation, MATSYA aims to deliver an adaptive, stealthy monitoring platform whose performance and ecological benefits are rigorously and experimentally quantified.

Keywords: Biomimetic robotics · Robotic fish · Computational fluid dynamics · Physics-informed neural networks · ROS 2 · Underwater monitoring

1 Introduction

Autonomous underwater robots are increasingly deployed for water-quality assessment, infrastructure inspection, habitat mapping, and biodiversity observation [5,7]. Executing these missions successfully requires platforms capable of navigating constrained environments, acquiring reliable sensor data, and operating with minimal ecological disturbance. While biomimetic, fish-inspired robots

present a promising pathway to agile and flexible locomotion, optimizing their performance requires careful integration of body geometry, actuator design, fin motion, and surrounding flow dynamics [2,4].

Recent research in robotic-fish hydrodynamics has advanced CFD-based optimization, highlighting the relationship between undulatory swimming behaviours and the cost of transport [2,3]. Concurrently, innovations in soft and wire-driven actuation demonstrate diverse strategies for achieving bio-inspired motion, while modular platforms have been conceptualized for sensor deployment [4,5,7]. Furthermore, PINNs offer a modern approach to fusing governing physical equations with empirical data, though their computational cost and convergence remain highly problem-dependent [1].

The primary challenge in this domain is bridging the gap between isolated component research and holistic system integration. MATSYA is proposed to resolve this fragmentation by linking hydrodynamic design, environmental sensing, and mission autonomy through a staged simulation-to-prototype workflow, ensuring that "low disturbance" is a validated metric rather than an assumed byproduct of biomimicry [2,3,4,1,5,7].

2 Research Gap and Objectives

Existing studies provide robust evidence for isolated subsystems—such as propulsion optimization, soft actuation, and physics-informed flow modelling—but do not demonstrate a unified underwater monitoring robot that guarantees efficient propulsion, autonomous operation, and measured ecological safety [2,3,4,1,5,7]. To address this gap, the MATSYA project outlines four primary objectives:

1. Develop a biomimetic propulsion and body/fin architecture optimized for underwater ecological monitoring [4,5].
2. Utilize CFD to evaluate hydrodynamic behaviour, systematically comparing fin-motion parameters and candidate geometries [2,3].
3. Investigate the viability of PINN-based surrogate models for accelerating flow predictions, maintaining rigorous cross-validation against experimental and CFD data [1].
4. Design a robust ROS 2 software architecture to connect state estimation, environmental sensing, closed-loop motion control, and mission execution.

3 System Architecture

MATSYA is structured as an integrated hardware and software ecosystem comprising five interacting layers.

3.1 Biomimetic Propulsion and Mechanical Design

The physical platform utilizes a streamlined, fish-like body equipped with an undulatory tail or actuated fins [4,5]. The definitive actuation mechanism will

be selected following a comparative analysis of mechanical complexity, sealing requirements, power consumption, controllability, and kinematic range, utilizing existing robotic-fish baselines for reference.

3.2 CFD and Hydrodynamic Modelling

High-fidelity CFD simulations will estimate the pressure and velocity fields surrounding candidate geometries [2,3]. Key design variables include swimming speed, body flexibility, oscillation frequency, phase, and fin amplitude. The analysis will output crucial performance indicators such as drag, thrust, wake structure, and cost of transport. To ensure reliability, all simulations will document boundary conditions and grid/time-step sensitivity, and will be corroborated by experimental measurements.

3.3 PINN-Assisted Flow Prediction

PINNs embed governing equations and measurement data into a machine learning objective, offering a potential mechanism for solving inverse or forward flow problems [1]. Within MATSYA, hybrid CFD-PINN models serve as a research exploratory tool rather than a guaranteed real-time replacement for CFD. PINN surrogates will be trained on sampled CFD data, evaluated on held-out conditions, and assessed based on inference time and prediction error. Should PINNs fail to meet runtime or accuracy thresholds, traditional reduced-order models or CFD-derived lookup tables will be utilized.

3.4 ROS 2, Sensing, and Autonomy

ROS 2 acts as the central middleware, bridging sensor drivers, motion commands, state estimation, and mission logic. The sensor payload will be tailored to the monitoring task and budget, potentially including depth sensors, inertial measurement units (IMUs), imaging sensors, and water-quality monitors. Additionally, proprioceptive sensors will track actuator states and robot motion to refine body response and fin position estimations.

3.5 Closed-Loop Control

A feedback controller will translate mission and navigation commands into precise fin-actuation signals, minimizing heading and trajectory errors. Initial phases will deploy transparent, baseline controllers (e.g., PID) prior to transitioning to advanced learning-based or adaptive paradigms. Physics-informed deep reinforcement learning presents a promising avenue for complex scenarios, such as swimming in vortex streets [6]. Controller robustness must be validated across diverse operating conditions and repeatable disturbances.

4 Methodology: Simulation to Prototype

The MATSYA development lifecycle is structured into five sequential phases.

4.1 Phase 1: Digital Model and Baseline Definition

Establish reference operating conditions, define propulsion kinematics, and create the initial digital geometry. Baseline configurations will be sourced from established robotic-fish literature [2,4,5]. Repeatable simulation cases, coordinate frames, and boundary conditions must be formalized prior to comparative testing.

4.2 Phase 2: Hydrodynamic Analysis and Design Optimization

Execute CFD simulations across a controlled matrix of geometry and fin-motion parameters [2,3]. Candidate designs will be evaluated based on wake characteristics, cost of transport, swimming speed, drag, and thrust. Upon prototype fabrication, simulation data will be strictly compared against physical tank-test measurements.

4.3 Phase 3: Surrogate-Model Assessment

To mitigate the computational burden of rapid parameter exploration, a PINN or hybrid surrogate will be trained for specific flow-prediction tasks [1]. Model viability will be determined by comparing predictions against held-out CFD data, documenting failure conditions, training cost, and error rates. PINN outputs will require independent validation before integration into safety-critical control loops.

4.4 Phase 4: ROS 2 Integration and Closed-Loop Testing

Integrate navigation functions, state estimation, sensor topics, and actuator commands within the ROS 2 environment. Testing will progress from teleoperated trials to autonomous trajectory/heading control, and ultimately obstacle avoidance. System performance will be quantified via tracking error, response time, data latency, and disturbance recovery.

4.5 Phase 5: Prototype and Environmental Validation

The fully assembled system will undergo controlled pool or tank testing prior to open-water field trials. MATSYA's performance will be benchmarked against defined controls, such as fixed-fin or rigid-body configurations utilizing basic controllers. Ecological impact claims will be substantiated through standardized measurements of wake-related disturbances, acoustic output, and, where feasible, approved biological response protocols.

5 Evaluation Metrics and Baselines

The evaluation framework relies on specific metrics across five domains. Thresholds will be established following initial prototype testing.

Table 1. Proposed metrics for staged evaluation of MATSYA.

Domain	Metrics	Comparison or Validation
Hydrodynamics	Thrust, drag, swimming speed, wake characteristics	CFD baseline; selected tank-test measurements
Efficiency	Electrical power, energy per distance, cost of transport	Compare geometries and fin-motion settings under matched conditions
Control	Heading error, trajectory error, response time, disturbance recovery	Baseline PID controller and repeatable disturbance cases
Autonomy	Navigation success, obstacle-avoidance rate, mission completion	Repeated test courses with documented initial conditions
Sensing	Measurement error, sensor availability, data latency	Calibration against suitable reference instruments
Disturbance	Acoustic output and wake-related measures; biological response if feasible and approved	Standardized protocol and matched operating conditions

Rigorous evaluation demands holding extraneous variables constant. Key baselines include: (i) conventional CFD versus PINN surrogates [1]; (ii) alternative propulsion configurations [4,5]; and (iii) interpretable baseline controllers versus advanced adaptive controllers [6].

6 Expected Contributions and Limitations

MATSYA delivers a structured methodological framework linking closed-loop underwater operation, software integration, hydrodynamic analysis, and biomimetic mechanical design. If successfully implemented, the platform will enable advanced water-quality monitoring, habitat mapping, and pollution surveys in ecologically sensitive zones demanding high manoeuvrability and minimal disturbance.

Currently, MATSYA operates as a proposed development framework rather than a fully validated robot. Expected engineering constraints include PINN training instability, CFD-to-hardware discrepancies, sensor drift, battery capacity limitations, actuator durability, and waterproofing challenges. Consequently, initial prototyping will prioritize a constrained set of testable functions, expanding into full autonomy only after foundational baselines are experimentally validated.

7 Conclusion

The MATSYA framework provides a comprehensive blueprint for developing a biomimetic underwater robot that synthesizes fish-inspired propulsion, ROS 2 integration, optional PINN-assisted flow modelling, and closed-loop control. The core research imperative is determining whether these disparate technologies can be unified into a reliable monitoring platform while strictly quantifying—rather than assuming—its ecological and energetic benefits. The proposed phased methodology, progressing from digital twins to physical prototypes, ensures a rigorous evaluation path. Future deployments must report performance against defined baselines, reserving claims of ecological superiority for platforms that provide concrete biological-response and disturbance data.

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